# DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

# EVALUATION SCHEME & SYLLABUS

# FOR

# B. TECH. THIRD YEAR

## Carpet & Textile Technology

# AS PER

# AICTE MODEL CURRICULUM

# [Effective from the Session: 2024-25]

## B.Tech. Second Year, Semester- V
## (Carpet & Textile Technology)

| | | | | | | | | Evaluation Scheme | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Sl. No.** | **Subject Codes** | **Subject Name** | **Type** | **Category** | **Periods** | | | **Sessional Component** | | **Sessional (SW) (TS/PS)** | **End Semester Examination (ESE)** | **Total** | **Credit** |
| | | | | | **L** | **T** | **P** | **CT** | **TA** | **CT+ TA** | **TE/PE** | | |
| 1 | BCT-501 | Carpet & Textile Testing - I | T | PCS | 3 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 4 |
| 2 | BCT-502 | Chemical Processing of Textiles - II | T | PCS | 3 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 4 |
| 3 | BCT-503 | Carpet Manufacture | T | PCS | 3 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 4 |
| 4 | BCT-051/ BCT-052 | Yarn Technology - III/ Fabric Technology - III | T | PEC | 3 | 0 | 0 | 20 | 10 | 30 | 70 | 100 | 3 |
| 5 | BCT-053/ BCT-054 | Textile Design Concept/Woven Textile Design | T | PEC | 3 | 0 | 0 | 20 | 10 | 30 | 70 | 100 | 3 |
| 6 | BCT-551 | Carpet & Textile Testing - I Lab | P | PCS | 0 | 0 | 2 | - | 50 | 50 | 50 | 100 | 1 |
| 7 | BCT-552 | Chemical Processing of Textiles - II Lab | P | PCS | 0 | 0 | 2 | - | 50 | 50 | 50 | 100 | 1 |
| 8 | BCT-553 | Carpet Manufacture Lab | P | PCS | 0 | 0 | 2 | - | 50 | 50 | 50 | 100 | 1 |
| 9 | BCT-554 | Internship Assessment /Mini Project/ Start-up and entrepreneurship* | P | | 0 | 0 | 2 | - | 100 | 100 | - | 100 | 2 |
| 10 | | HSMC-II Audit Group/ HSMC-I Audit-Group | T | HSMC | 2 | 0 | 0 | 20 | 10 | 30 | 70 | 100 | 0 |
| | | **Total** | | | **17** | **3** | **8** | | | | | | **23** |
| | | **Minor Degree/Honors Degree MT-2/HT-2** | | | | | | | | | | | |

***** The Mini Project or internship or start-up and entrepreneurship (4-6 weeks) conducted during summer break after IV semester and will be assessed during V semester.

## B.Tech. Second Year, Semester- VI

## (Carpet & Textile Technology)

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Evaluation Scheme | | | | | |
| Sl. No. | Subject Codes | Subject Name | Type | Category | Periods | | | Sessional Component | | Sessional (SW) (TS/PS) | End Semester Examination (ESE) | Total | Credit |
| | | | | | L | T | P | CT | TA | CT+ TA | TE/PE | | |
| 1 | BCT-601 | Carpet & Textile Testing - II | T | PCS | 3 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 4 |
| 2 | BCT-602 | Structural Properties of Fibres | T | PCS | 3 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 4 |
| 3 | BCT-603 | Home Textile Technology | T | PCS | 3 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 4 |
| 4 | BCT-061/ BCT-062/ BCT-063 | Chemical Processing of Textiles - III/Knitting and Nonwovens/Fibres for Carpets & Textiles | T | PES | 3 | 0 | 0 | 20 | 10 | 30 | 70 | 100 | 3 |
| 5 | | Open Elective- I | T | OES | 3 | 0 | 0 | 20 | 10 | 30 | 70 | 100 | 3 |
| 6 | BCT-651 | Carpet & Textile Testing - II Lab | P | PCS | 0 | 0 | 2 | - | 50 | 50 | 50 | 100 | 1 |
| 7 | BCT-652 | Carpet & Textile Design Lab | P | PCS | 0 | 0 | 2 | - | 50 | 50 | 50 | 100 | 1 |
| 8 | BCT-653 | Home Textile Technology Lab | P | PCS | 0 | 0 | 2 | - | 50 | 50 | 50 | 100 | 1 |
| 9 | | HSMC-II Audit Group/ HSMC-I Audit-Group | T | HSMC | 2 | 0 | 0 | 20 | 10 | 30 | 70 | 100 | 0 |
| | | **Total** | | | **17** | **3** | **6** | | | | | | **21** |
| | | **Minor Degree/Honors Degree MT-3/HT-4** | | | | | | | | | | | |

***** The Mini Project or internship or start-up and entrepreneurship (4-6 weeks) conducted during summer break after VI semester and will be assessed during VII semester.

# SYLLABUS

# BCT501 Carpet & Textile Testing –I

**Course Objectives:**

To learn the importance of testing, role of environmental conditions on testing results, measurement of fibre length, fibre fineness, fibre maturity, yarn twist, yarn irregularity and their effects on end product quality. Also, to get an understanding of the terminology related to construction and testing of carpets and to acquire knowledge on fire testing of carpets.

**Unit (1): Total lectures required = 8**

Objectives of Testing, selection of samples for testing, Random and biased sampling, Principles of sampling for fibre/yarn/fabric/carpet testing, Effect of moisture on different fibre properties, moisture content & regain; measurement of atmospheric conditions, Brief description and working principle of moisture meter.

**Unit (2): Total lectures required = 8**

Importance of fibre length measurement, Concept of Baer Sorter Diagram, Analysis of effective length, modal length, Average length and short fibre% using Baer Sorter Diagram, Concept of span length and Uniformity ratio. Importance of fibre fineness measurement, Determination of fibre fineness using air flow method, Fibre maturity: methods of determination of fibre maturity. Salient features of modern testing instruments like HVI, AFIS, WIRA FDM.

**Unit (3): Total lectures required = 8**

Importance of yarn Twist & its measurement, Measurement of yarn hairiness, Type of irregularity and its expression; Index of irregularity; limit irregularity; Principles of measurement of Evenness Testing (Sliver, Rove & Yarn), nature of irregularity; periodic irregularity, random irregularity.

**Unit (4): Total lectures required = 8**

Terminology related to construction and testing of carpets, Measurement of carpet thickness, Pile height, Knots per square inch, Tuft density, Carpet wear & abrasion, Thickness & compression recovery, Tuft Withdrawal force etc.

**Unit (5): Total lectures required = 8**

Terms & definitions related to Fire Testing of carpets, Factors affecting behavior of carpet against flame, Various Carpet Flammability Testers

**References**:


1. Principles of Textile Testing by J. E. Booth
2. Physical Testing of Textiles by Saville
3. Handbook of Textile Testing & Quality Control by Grover & Hamby
4. Physical Testing & Quality Control by K. Slater (Textile Institute)
5. Testing & Quality Management by V. K. Kothari
6. Various National & International testing standards related to carpet & textiles


**Course outcome:**

After completion of this course, students will be able to:

1. Appreciate the importance of Carpet and Textile testing.
2. Identify the principles used for sampling of fibre/yarn/fabric/carpet during their testing
3. Appreciate the importance of environmental condition and its measurement techniques
4. Demonstrate the testing of length, fineness and maturity of fibre.
5. Understand the causes of yarn hairiness, yarn irregularities and their measurements.
6. Understand the terminology related to construction and testing of carpets.
7. Implement the knowledge of Fire Testing of carpets to design the product accordingly.

# BCT502 Chemical Processing of Textiles II

**Course Objectives:**

Objective of the course is to learn, understand, identify dyeing process & finishing process and apply dyes: azoic dyes, sulphur dyes, vat dyes, reactive dyes, disperse dyes to textiles and apply finishing chemicals to textiles.

**Unit (1): Total lectures required = 8**

**Azoic Colour:** Basic principle of application of Azoic Colour; Naphtholation process, role of additives and effect of process parameters on naphtholation process. Classification of naphthols based on substantivity. Basic chemistry in Diazotisation process, Procedure for diazotisation; Stabilization of diazonism salts. Development of Shades- Coupling, Basic Chemistry involved in coupling. Fastness properties of Azoic Colours.

**Sulphur Dyes:** Chemical nature of sulphur dyes; Classification of sulphur dyes, Process of application of sulphur dyes & mechanism of dyeing, Role of additives and process parameters in sulphur dyeing. Fastness properties of sulphur dyes.

**Unit (2): Total lectures required = 8**

**Vat Dyes:** General formula of vat dyes, classification of vat dyes; Process of application of vat dyes, mechanism of vat dyeing; oxidation process; Role of additives and process parameters in vat dyeing.

Sollubilised Vat dyes. Dyeing with solubilizsed vat dyes. Fastness properties of vat dyes.

**Unit (3): Total lectures required = 8**

**Reactive Dyes:** General formula of reactive dyes. Chemical nature of reactive dyes; Classification of reactive dyes; Process of application of various kind of reactive dyes & dyeing mechanism; Role of additive and effect of process parameter in reactive dyeing. General properties of reactive dyes.

**Disperse Dyes:** Chemical nature of disperse dyes, application of disperse dyes on polyester; dyeing mechanism; Role of additives & effect of process parameters on disperse dyeing. Optical whitening agents, application to polyester and cotton.

**Unit (4): Total lectures required = 8**

**TEXTILE FINISHING**

Purpose of finishing; Type of finishing, Permanent & Temporary; Mechanical & Chemical finishing. Chemical finishing- Anticrease finish (Using DMU & DMDHEU)

**Unit (5): Total lectures required = 8**

Flame retardant finish, Water repellent & Water proofing finish, Softening of fabric, Anti soiling finish.

Brief description of different type of finishing machine; Stenter, Sanforizing machine, Calendering machine etc.

**References:**


1. Dyeing & Chemical Technology of Textile fibres by E.R. Trotman.
2. Chemical Technology of fibrous material by F..Sadov.
3. Chemistry of Dyes and principles of Dyeing by Dr. V.A. Shenai.
4. Technology of Dyeing by Dr. V.A. Shenai
5. Chemical Processing of Synthetic fibres by Dr. K.V. Datye & A.A. Vaidya
6. The Dyeing of Cellulose fibres by Clifford Preston.
7. Technology of Finishing – Vol. X by Dr. V.A. Shenai.
8. Introduction to Textile finishing by J.T. Marsh


**Course Outcomes:**

At the end of the course students will be able to:

1. Knowledge of fundamentals of azoic dyes, sulphur dyes, vat dyes, reactive dyes, disperse dyes and textile finishing.

2. Understanding of technological involvement with appropriate techniques in azoic dyes, sulphur dyes, vat dyes, reactive dyes, disperse dyes and textile finishing.
3. Knowledge of application technique of azoic dyes, sulphur dyes, vat dyes, reactive dyes, disperse dyes and textile finishing process in industry.
4. Capacity to make the chemical processes under control in azoic dyes, sulphur dyes, vat dyes, reactive dyes, disperse dyes and textile finishing.
5. Understanding for doing project work and aptitude for higher studies in the field of azoic dyes, sulphur dyes, vat dyes, reactive dyes, disperse dyes and textile finishing.

# BCT503 Carpet Manufacture

**Course objectives:**

A comprehensive understanding of various types of handmade and machine made carpets, their manufacturing processes and quality control measures.

**Unit (1):** **Total lectures required = 8**

Classification of hand made carpets and brief description of each type. Yarn used in hand made carpet.

Brief introduction of hand made carpet sector of India and abroad. Carpet export proceedings and process control of hand made carpet sector.

Designing of hand made carpet- brief introduction of CAD for carpet.

**Unit (2):** **Total lectures required = 8**

Hand Knotted Carpet:

Terms and definitions, Persian, Turkish, tibetan and other type of knot structure and its quality parameters.

Manufacturing of various hand knotted carpets - Preparatory process, tools and equipments used for Persian, Tibetan, shaggy carpets.

Washing and other mechanical finishing for hand knotted carpets.

Hand made dhurry:

Techniques , Quality and other process parameter of Hand made flat carpet on horizontal / Vertical loom frame viz. panja dhurry, Shoumak , Chindi dhurry and other fancy products.

Defects in hand knotted carpets and their remedies.

**Unit (3):** **Total lectures required = 8**

Hand loom Carpets:

Flat carpet making – quality parameters, jacquard dhurry making and its finishing.

Pile carpet making - Preparatory process and weaving techniques for plain and Jacquard loom pile carpet.

**Unit (4):** **Total lectures required = 8**

Tufted Carpets:

Structure of tufted carpet, backing cloth and quality parameters.

Hand tufting process - Preparatory process, tools and equipments , tufting frame, manual / electric hand tufting gun

Backing of hand made carpet, Natural latex, synthetic resin and its composition of backing solution for hand made carpet.

Finishing of hand tufted carpet.

Defects in hand made carpets and their remedies.

**Unit (5):** **Total lectures required = 8**

Introduction to machine made carpet.Brief introduction to Broad loom woven,tufted carpets and other types of carpet machines.

**References:**

1. Advances in Carpet Manufacture by K. K. Goswami, Woodhead Publishing
2. Journals & Magazines 3. Carpet-e-World
3. Carpet Manufacture by Crawshaw
4. Tufted Carpet by Von Moody
5. Process control in carpet manufacturing by K KGoswami, Abhishek Publishing, Chandigarh, India.

**Course outcomes:**

After completing this course, the student will be able to

1. List different types of carpets, create designing of carpets through CAD, understand terms used in the manufacture and procedures involved in the export of carpets
2. Analyze the difference between different types of knots, explain tools and equipments used in all types of manufacture, preparatory process, washing and finishing of carpets and defects in the carpet
3. Demonstrate the preparatory process, quality parameters, and manufacturing of handloom carpets
4. Differentiate between the manufacture of hand tufted carpets with knotted carpets , finishing of tufted carpets with different types of latex, defects and remedies
5. Classify and understand the machine made carpets

# BCT-051 Yarn Technology-III

**Course Objectives:**

To learn, demonstrate, evaluate and product development on sustainable modern techniques of advance yarn manufacturing namely open end spinning, such as rotor, friction, airjet, air vortex, electrostatic, twist less, plyfill etc. and woolen and semi-worsted yarn formation technology.

**Unit (1):** **Total lectures required = 8**

Various systems of wool fibre spinning- woolen, worsted, semi worsted - Flow chart, Willow m/c, woolen and semi worsted cards, production calculations in woolen card.

**Unit (2):** **Total lectures required = 8**

Gilling, Rubbing frame and ring frame-sequence of machines and operations, Comparison of above spinning systems, production calculations in these machines.

**Unit (3):** **Total lectures required = 8**

Rotor Spinning: Principle of Operation, properties and applications of rotor yarn.
Friction Spinning: Principle of operation of DREF-II & DREF-III, properties and application of friction spun yarn

**Unit (4):** **Total lectures required = 8**

Air Jet Spinning: Principle of formation of yarn in Murata Air jet spinning, Properties of Air jet spun yarn & the factor influencing air jet spun yarn.

**Unit (5):** **Total lectures required = 8**

Brief discussion on Ply-fill spinning system, wrap spinning, electrostatic spinning etc. Requirement of carpet yarn in regards to count, twist, and bulk for Handmade and machine made carpet. Faults in carpet yarn and their remedies.

**References**:


1. New spinning technology- W. Klein.
2. Worsted yarn technology-By Eric Oxtoby
3. C. Vickerman, Woollen Spinning, Abhishek Publications, Chandigarh-17 (India)
4. W S Simpson and G H Crawshaw, Wool: Science and Technology, Woodhead Publishing Limited, Cambridge, England
5. Miles Collins, Wollen and Worsted Spinning, Abhishek Publications, Chandigarh-17 (India)


**Course Outcomes**:

At the end of the course students will be able to

1. Identify various advance spinning techniques for advance yarn manufacturing.
2. Classify technology of modern spinning machines such as rotor, friction, airjet, air vortex, electrostatic, twist less, plyfill etc.
3. Demonstrate various modern techniques of spinning as in CO 1 & CO 2.
4. Compare and contrast various modern woolen and semiworsted techniques.
5. Review and revise the knowledge of various spinning and other techniques of un conventional yarn formation techniques.
6. Implement the knowledge and design innovative techniques for advance yarn manufacturing on above techniques.

# BCT-052 Fabric Technology III

**Course Objectives:**

To learn, demonstrate, evaluate and product development on sustainable modern techniques of weaving namely automatic loom, shuttles less weaving such as gripper, and jet looms, latest techniques of multiphase and triaxial weaving, terry and leno fabric technology.

**Unit (1):** **Total lectures required = 8**

Automatic shuttle loom:

Features of automatic shuttle looms, type of automatic shuttle looms, Automatic cop change loom mechanism. Weft feeler and transfer mechansim, Mechanism of shuttle change looms, Introduction to bobbin loader and unifil loom winders.

**Unit (2):** **Total lectures required = 8**

Shuttle less weaving:

Advantages, Essential features - classification of shuttleless picking system, let off and take mechanism, matched cam beat up mechanism, Types of selvedges- tucked, leno selvedges.

**Unit (3):** **Total lectures required = 8**

Gripper loom:

Projectile Picking cycle, projectile, accumulator, and other essential parts of picking cycle. picking mechanism, technical specification of modern machines.

Rapier picking system - Classification of Rapier and brief description of each type, rapier head and Picking cycle of Dewas and gabler picking system, rapier drive. Technical specification of Modern machines.

**Unit (4):** **Total lectures required = 8**

Jet Loom:

Air jet picking system, Requirement of air, Picking cycle - confusor, Nozzel, PMD, waste selvedge, suction device. Technical specification of Modern machines.

Water jet picking system, Requirement of water and pump, picking cycle, Nozzle, suction device. Technical specification of Modern machines.

**Unit (5):** **Total lectures required = 8**

Multiphase weaving: shedding, picking and beat up of M8300 loom,

Triaxial weaving principle and circular weaving.

Terry pile weaving – essential requirements, various terry mechanisms – loose reed, cloth control and variable sley sweep based terry mechanism, pile height adjustment.

Principle of Leno weaving – classes of leno structures, string and steel doup mechanisms.

**References**:


1. Principles of weaving by Marks & Robinson
2. Weaving Mechanism by T. Fox
3. Weaving machine and mechanism by Talukdar, Azgaonkar and Sriramulu
4. Fancy Weaving By K T Ashwani.
5. Woven Fabric Production – I, II NCUTE Publications
6. ADAVANCETextile Design W Watson
7. Weaving machine, mechanism, management by Talukdar, Sriramulu, Ajgaonkar
8. Shuttleless Weaving Swaty
9. Weaving Mechanism, Part I, II by N N Banerjee
10. Publications on Weaving Mechanism Prof. G. S. Bhargava
11. Manual of Manufacturing Company SULZER, NISSAN

**Course outcome:**

At the end of the course students will be able to:

1. Identify various advance weaving techniques for fabric manufacturing.
2. Classify technology of modern weaving machines such as automatic gripper, jet and multiphase weaving, Terry, leno and tri axial weaving.
3. Demonstrate various modern techniques of weaving as in CO 1 & CO 2.
4. Compare and contrast various modern weaving techniques.
5. Review and revise the knowledge of various shuttle less weaving and other techniques of unconventional weaving techniques.
6. Implement the knowledge and design innovative techniques for weaving on above techniques.

# BCT-053 Textile Design Concept

**Course Objective:**

To develop knowledge of traditional, modern, abstract, contemporary design and sustainably reproduce various carpet and textile design with the help of colour forecast and color matching tools.

**Unit (1): Total lectures required = 8**

**Principle of Design**

Elements of textile design, Basic of carpet designs creations, commercial aspects of design with Basic Elements and Market Trends, Material and method, Drawing from outdoor sources i.e. Park, Museum, Architectural Building, ,

**Unit (2): Total lectures required = 8**

**Design elements from Textile Design**

Kantha, Chikenkari, Bagh Phulkari, Banarasi Brocades, Indian Motif (Indiya Collection); Design & Drawing – Historical, Aesthetical, Traditional Textile etc.

**Unit (3): Total lectures required = 8**

**Design Concept from Motif & Printed Design**

Jamdani, Bandhani (Tie & Dye)Bagroo, Sanganer, Batik Prints, Digital Printing, Screen Printing, Block Printing, Colour forecasting of Carpet & Textile Design,

**Unit (4): Total lectures required = 8**

**Brief discussion on traditional carpet & floor covering**

Mir, Prayer rug, Abusson, Herati, Isfahan, Kirman, Kazak, Heriz, Kashan, Saroukh, Bidjar, Tabriz etc, Tufted,Tibbettan, Dhurries (Geometrical & Floral) – Killims, Different quality calculation of Rugs.

**Unit (5): Total lectures required = 8**

**Use of Motifs in Carpet, Textile Design with Advance Features**

Drawing of all designs studied in this module & creating different types of design, Modern look, traditional look of design by mixing/modifying different design styles, Introduction of Computer Aided Design (CAD), Importance & Advantage of CAD.

**References:**


1. Traditional Needle Arts Embroidery by Katrin Cargill, Great Britain.
2. Indian Embroidery by Kamala Devi Chartophadhya, Wiley Einstein Ltd., Delhi.
3. Saris of India by RTA Kapur& Amba Sanyal,Wiley Eastern Ltd., New Delhi
4. Hand book of textile design Jacquir Wilson, wood head publishing, UK.
5. Watson's Textile Design and Colour by Z Grosicki; Universal Publishing Corporation, Bombay (India)
6. Textile Design by Thames & Hudson
7. Persian Carpets by Dr. Seyed
8. Hand Crafted Indian Textile by Roli Books
9. Heritage by Design Point
10. Carpet Style by Phillips, Barty
11. Carpet Manufacture by Crawshaw, G.H.
12. Carpets and Textiles by Spuhler, Friedrich
13. Carpets: Techniques, Traditions and History by Anquetil, Jacques
14. History of Textile Design by Shenai,V.A.
15. The Indian Textile Journal (ITJ) [Periodical].
16. Ancient Indian Textile Designs - Part – I by Mishra, Jai Shankar

17. Positive Design - I Flower by Shoin, Kyoto
18. Rugs & Carpets from the Collection of Dildarian by Sotheby's
19. Floral patterns by Roojen, P.V.
20. Carpet and Textile Patterns by Purdon, Nicholas
21. Advance carpet Manufacturing, K K. goswami, wood head publisher UK.
22. Nedgraphics Texcelle Software Manual
23. Indian Carpet & Textile Software Manual

**Course outcomes:**

At the end of the course students will be able to:

1. Describe knowledge of design element which will help in textile Design creations.
2. Illustration of different type of textile, carpet, motif, composition through CAD.
3. Explanation of different type of modern and traditional design and printing knowledge will be easy to connect with new fashion.
4. Depict traditional knowledge of carpet can help in new innovation of design.
5. Represent knowledge color trend year wise combination, texture, different character of colour shade, which will help in market representation.
6. Explanation of different character according to buyers like Modern, Abstract, Contemporary Look, Traditional Modify different character.

# BCT-054 Woven Textile Design

**Course Objective:**

To learn, demonstrate, evaluate and develop basic woven designs (like plain, satin, mock leno, diamond and their derivatives) and advanced woven designs (like, extra thread designs, backed fabric, double cloth, treble cloth and jacquard design) and also sustainably development of each design.

**Unit (1): Total lectures required = 8**

Introduction to Elements of woven designs, Classification of woven fabric, concept of design, draft, peg plan, denting order. plain weave, derivatives of plain weave and different methods of decorating of plain fabric

**Unit (2): Total lectures required = 8**

Classification of twill weaves. Elongated, Zig-zag, wavy, broken, herringbone, transposed, Rice, & rearranged twill weaves. Satin /Sateen weave.

Mockleno, diaper, diamond, Cork-Screw & crepe weaves . Different methods of producing crepe weaves; Towelling weaves: diamond, mockleno, huck a back, honey comb

**Unit (3): Total lectures required = 8**

Principle and objective of extra warp and extra weft designing loom requirements, methods of disposing surplus thread, difference between extra warp and extra weft designing

Principle and objective of warp backed and weft backed fabric. Loom requirements, examples of each category, difference between warp backed and weft backed fabric.

**Unit (4): Total lectures required = 8**

Double cloths: Introduction, classification, self stitched double cloths, centre stitched double cloth, interchanging double cloths. Example of each type. Introduction of Treble cloth.

**Unit (5): Total lectures required = 8**

Development of jacquard design, associated process and calculation. Punching card in piano card cutting machine, card lacing etc.

**References:**


1. Watson's Textile Design and Colour by Z Grosicki; Universal Publishing Corporation, Bombay (India)
2. Watson's Advance Textile Design
3. Grammer of Textile Design – Nisbet
4. Structural Fabric Design by – Kilby
5. Woven Structures and Design – Doris Goerner; British Textile Technology GroupWIRA, House, Leeds (UK)

**6.** Fibre to Fabric by Ghosh


**Course outcome:**

At the end of the course students will be able to:

1. Describe the knowledge in Basic weave, Plain, Twill, Satin, toweling weave, Extra thread design, backed fabric design, double and treble cloth and producing design on jacquard weaving.
2. Classify and identify Basic weave, Plain, Twill, Satin, toweling weave, Extra thread design, backed fabric design, double and treble cloth and design on jacquard weaving.
3. Sketch Basic weave, Plain, Twill, Satin, toweling weave, Extra thread design, backed fabric design, double and treble cloth and jacquard design.
4. Solve and compare Basic weave, Plain, Twill, Satin, toweling weave, Extra thread design, backed fabric design, double and treble cloth and jacquard design.
5. Predict and recommend (on the basis of end use requirement) Basic weave, Plain, Twill, Satin, toweling weave, Extra thread design, backed fabric design, double and treble cloth and jacquard design.

6. Create and develop Basic weave, Plain, Twill, Satin, toweling weave, Extra thread design, backed fabric design, double and treble cloth and jacquard design.

# LAB 1: BCT551 Carpet & Textile Testing - I Lab

**Course Objectives:**

1. To get aware of principles of measurement of environmental conditions of testing lab.
2. To develop an understanding of yarn count system using testing instruments.
3. To get an understanding of the principles of measurement of fibre length and fibre fineness.
4. To acquire knowledge on testing of basic Capet constructional parameters.
5. To acquire knowledge on Capet flammability testing.

**Experiments:**

1. Measurement of Relative humidity of the testing laboratory.
2. Determination of count of a given yarn/sliver/roving in different counting system.
3. To study the principle of measurement of fibre length of cotton by means of Digital Fibrograph and determine:- a. 2.5 % Span Length b. 50 % Span Length c. Uniformity.
4. Measurement of fibre length of wool fibre by means of W.I.R.A FDA.
5. Determination of fineness of wool fibre by WIRA tester.
6. Determination of fineness and maturity of Cotton fibre.
7. Determination of % medullation of wool fibre using projection microscope.
8. Determination of twist in single and folded yarn and to determine twist factor used in spinning the yarn.
9. Measurement of pile height of carpet using leaf gauge.
10. Determination of thickness of carpet using portable thickness gauge.
11. Determination of tuft withdrawal force using tuft withdrawal tensometer.
12. Determination of surface flammability of carpet by 16 CFR 1630/1631 method.
13. Determination of flammability of carpet by BS 4790 method.

NOTE:

Experiments shall be decided on factors like: • Facilities installed at Institute. • Accessibility to Industry & nearby Institutes. • Trend of Technological Developments in National & International perspective.

**Course outcome:**

After completion of this course, students will be able to:

1. Appreciate the importance of measurement of environmental conditions of testing labs.
2. Measure the fibre length and fibre fineness using various instruments.
3. Measure the count of Yarn/roving/sliver using various instruments.
4. Understand the basic testing parameters related to construction and Capet flammability.

# LAB 2: BCT552 Chemical Processing of Textiles - II Lab

**Course Objectives:**

1. To synthesize and apply azo dyes by coupling reaction.
2. To apply vat dyes on cotton fabric/yarn.
3. To dye cotton fabric/yarn with remazol & procion dyes and sulphur dyes.
4. To operate HTHP machine and apply disperse dyes on polyester fabric/yarn.
5. To determine washing fastness of dyed textiles by standard (IS/ISO) method.
6. To determine light fastness of dyed textiles by standard (IS/ISO) method.

**Experiments:**

1. Development of dye by coupling method.
2. Dyeing of cotton yarn with vat dyes in sample pot dyeing machine.
3. Dyeing of cotton yarn with sulphur dyes
4. Dyeing of cotton yarn with Remazol dyes.
5. Dyeing of cotton yarn with Procion Dyes
6. Dyeing of cotton yarn with Bi-functional reactive Dyes
7. Dyeing of polyester yarn/fibre in laboratory HTHP machine.
8. Application of optical whitening agent on cotton.
9. Determination of washing fastness of dyed material.
10. Determination of washing fastness of dyed material.

**Course Outcomes:**

At the end of the course students will be able to:

1. Develop azo dye by coupling method.
2. Apply vat dyes on cotton fabric/yarn.
3. Dye cotton fabric/yarn with remazol & procion dyes and sulphur dyes.
4. Operate HTHP machine and apply disperse dyes on polyester fabric/yarn.
5. Evaluate washing fastness of dyed textiles by standard (IS/ISO) method.
6. Evaluate light fastness of dyed textiles by standard (IS/ISO) method

# LAB 3: BCT553 Carpet Manufacture Lab

**Course objectives:**

1. To get knowledge about various types of carpets and raw materials used in manufacture
2. To acquire knowledge about dyeing of yarn used in carpets
3. To give an in depth knowledge about knotting and tufting of carpets
4. To acquire knowledge about machine made carpets
5. To develop an understanding on carpet washing and finishing
6. To familiarize about the testing equipments.
7. To get knowledge about data collection

**Experiments:**

1. Identify various types of carpet and raw material used
2. Seminar on carpet designing and quality of hand knotted carpet
3. Demonstration of carpet yarn dyeing
4. Practice of hand knotted and tufted carpet manufacturing.
5. Demonstration on machine made carpet
6. Demonstration of carpet washing and finishing
7. Demonstration of various carpet testing instruments in the lab.
8. Data collection of carpet export and report making

**Course outcomes:**

After completing this course, the student will be able to

1. Understand about various types of carpets and raw materials used in manufacture
2. Understand the dyeing of yarn used in carpets
3. Understand the knotting and tufting of carpets
4. Understand the carpet washing and finishing
5. Understand the machine made carpets
6. Understand about the equipments used for testing and assessing the quality

# BCT601 Carpet & Textile Testing –II

**Course Objectives:**

To learn the principles involved in mechanical strength testing of textiles, importance of testing of functional properties as well as serviceability aspects of carpets and fabrics.

**Unit (1): Total lectures required = 8**

Terms and definitions used in mechanical strength testing of textiles, Factors affecting the tensile properties of yarn, Classification of tensile strength testing machines based on operating principles (CRE, CRL & CRT): Brief Description and principle involved in working of a) Lea Strength tester b) Tensile Tester operating on Strain Gauge principle. Fabric tensile testing: Sample preparation – Strip Test & Grab Test, Fabric Tear and bursting strength testing

**Unit (2): Total lectures required = 8**

Terms & definitions related to Air permeability of fabric, Measurement of Air permeability using Shirley Air Permeability Tester, Brief idea on thermal properties of fabric. Terms & definitions related to Water proofing property of textile materials, Methods of testing: (i) Spray Test ii) Bundesman's Water repellency test iii) Drop penetration test, fabric wetting property, Wicking property, waterproof breathable fabrics. Flammability testing: Terms and definitions related to flammability of textiles; Determination of flammability of fabric.

**Unit (3): Total lectures required = 8**

Serviceability, Wear and abrasion resistance testing: Principles of testing of abrasion resistance testing; factors affecting abrasion resistance: Brief description of Martindale Abrasion Tester: Pilling Tests: Assessment of pilling of fabric using ICI pilling box. Salient features of FAST system, KESF system.

**Unit (4): Total lectures required = 8**

Brief description and operating principle of following carpet testing equipment's like Dynamic loading machine, Carpet Static Loading Device, Pilfuz Carpet Tester, Usometer, Hexapod Tumble Tester, Roller Chair Testing Device.

**Unit (5): Total lectures required = 8**

Relevance of testing of functional properties of carpets like – Appearance retention, Carpet durability including soil ability, other properties like insulation, acoustic, electrostatic properties etc.,

**References:**


1. Principles of Textile Testing by J. E. Booth
2. Physical Testing of Textiles by Saville
3. Handbook of Textile Testing & Quality Control by Grover & Hamby
4. Physical Testing & Quality Control by K. Slater (Textile Institute)
5. Testing & Quality Management by V. K. Kothari
6. Various National & International testing standards related to carpet & textiles


**Course outcome:**

After completion of this course, students will be able to:

1. Classify the instruments used for mechanical strength testing of textile materials.
2. Demonstrate the measurement of working life/ durability of carpet and textile articles.
3. Implement the knowledge of various functional properties of carpet and textiles to design the product accordingly.

# BCT602 Structural Properties of Fibres

**Course Objectives:**

To learn the importance of morphological structure on various electrical, optical, frictional and mechanical properties and moisture absorption behaviour of textile materials. Also, to get an understanding of the concept of various thermal transitions occurring in fibres at molecular level and their characterization techniques.

**Unit (1):** **Total lectures required = 8**

Amorphous and crystalline phases, molecular chain length, Glass Transition, Plasticization, Crystallization, Melting, factors affecting Tg & Tm, Role of molecular entanglement on fibre formation.

**Unit (2):** **Total lectures required = 8**

Differential heat of sorption, integral heat of sorption, Moisture absorption, effect of hydrophilic groups, moisture absorption in crystalline and non crystalline region, directly and indirectly attached water, creasing behaviour in textiles.

**Unit (3):** **Total lectures required = 8**

Basic definitions related to mechanical properties of textiles (tenacity elongation, modulus, work of rupture, Elastic recovery), Explanation of load-elongation and stress-strain curves related to textile fires, Relation between structure and mechanical properties of fibres, time effects (Primary and secondary creep).

**Unit (4):** **Total lectures required = 8**

Thermal behaviour of textile fibres by DSC, TGA, Thermal Mechanical Analysis, Density Gradient Column, Preparation of density gradient column.

**Unit (5):** **Total lectures required = 8**

Optical properties of fibres, Birefringence behaviour, dielectric properties, fibre friction, fibre friction measurement and static charge measurement.

**References:**


1. Manufactured fibre technology by V.B. Gupta, V.K. Kothari
2. Physical properties of fibre by J.W.S. Hearle
3. Textile fibres by ATIRA
4. Principles of Textile Testing by J. E. Booth
5. Physical Testing of Textiles by Saville


**Course outcome:**

After completion of this course, students will be able to:

1. Understand the importance of crystalline region, amorphous region, molecular chain entanglements etc. with respect to the morphological development of textile fibres.
2. Judge the role of morphological structure in moisture absorption behaviour of textile materials.
3. Appreciate the importance of morphological structure in development of mechanical properties in textile materials.
4. Understand the principles involved in characterization of various electrical, optical and frictional properties of fibres.
5. Explain the working principle of various sophisticated testing instruments such as DSC, TGA, DGC, DMA etc.

# BCT603 Home Textile Technology

**Course Objective**

To learn, demonstrate, evaluate and product development on sustainable manufacturing and embellishment techniques of home textiles namely marker planning, spreading, cutting, sewing, embroidery, patchwork, appliqué and quilting.

**Unit (1): Total lectures required = 8**

Introduction to Home Textiles, areas of application, the planning, drawing and reproduction of the marker, the requirements of marker planning, efficiency of the marker plan, methods of marker planning and marker use. the spreading of the fabric to from a lay, the requirement of the spreading process. methods of spreading, the nature of fabric packages. the cutting of fabric: objectives of cutting, cutting machines: hand shears, straight knife, round knife, band knife etc.

**Unit (2): Total lectures required = 8**

Stitch, stitch types, sewing machine-parts and their function, sewing machine feed mechanisms, seam, seam types, seam finishes.

Sewing machine needles: functions, parts, selection of sewing needle.

Sewing threads: fibre type, construction, thread sizing and thread packages.

Associated work aids of sewing machines.

**Unit (3): Total lectures required = 8**

**Fullness:** Definition, darts, tucks, pleats. flares, godets, gathers, shirrs and frills or ruffles.

**Placket finishes:** Definition, classification continuous bound, bound and faced (two piece) plackets, zipper placket. tailored placket.

**Embroidery**: Introduction, different types of embroidery stitches, different materials/ equipments used in embroidery, precautions during setting the fabric on embroidery frame, darning.

**Unit (4): Total lectures required = 8**

**Patch Work:** Definition, different styles of patchwork techniques such as pieced patchwork, shell patchwork, suffolk puffs, crazy patchwork, log cabin patchwork, strip patchwork, seminole patchwork, folded star patchwork, mayflower patchwork and pleated patchwork.

**Appliqué:** Definition, various styles of appliqué techniques, standard appliqué, appliqué perse, reverse appliqué, padded appliqué, folded appliqué, shadow appliqué, lace appliqué

**Quilting:** Definition, various styles of quilting, wadded quilting, padded quilting, corded quilting, shadow quilting.

**Unit (5): Total lectures required = 8**

**Curtains**: - Definition, various styles, choices of fabrics, calculating the amount of material needed, manufacturing steps of curtains.

**Bedspreads:** Estimating yardage for a spread, bedspread types and its manufacturing: throw, flounced and tailored.

**Pillows:** Types of pillows, knife-edge pillow, box-edge pillow, basic measurements, constructing a knife-edge pillow covers, tufting pillows, pillow shams, ruffled pillow, pillow sham with flat self-border, box-edge pillow cover: rectangular box-edge cover, boxed effect without boxing strip, circular box-edge cover, bolster and its covers: round bolster and wedge bolster.

**Reference:**


1. Carr & Latham's technology of clothing manufacture revised by David J. Tyler (Fourth edition), Blackwell publishing.
2. Introduction to clothing production management-second edition. A.J.Chuter, Blackwell
3. science, New Delhi.

4. Garment technology for fashion designers-Gerry Cooklin, Blackwell science, New Delhi.
5. Introduction to clothing manufacture- Gerry Cooklin, Blackwell science, New Delhi.
6. Process control in home textiles manufacturing by K K Goswami, abhishek publishing
7. Chandigarh, India.
8. Soft furnishing book by Kartin Cargill, Reed consumer books limited, London.
9. Soft furnishing by saarah Campbell and Hilary More, MacDonald books, QED publishers
10. Limited London.
11. Simplicity's (1993) –Simply the best home decoration book, A fire side book as published by Simon and Schulster (New York). London. The simplicity pattern company Inc.
12. Home Fashion
13. Cloths line (Journal)
14. House & Garden (Journal)
15. Textiles Para El Hogar (Journal) Distribution & Suibscription – Ecuadar, 75, entresuelo,
16. 08029 Barcelona, Espane, e.mail: publica@publica.es, castellon@publica.es
17. Process control in home textiles manufacturing K K Goswami, Abhishek publishing, Chandigarh.

**Course outcome:**

At the end of this course students will be able to:

1. Define home textiles and identify its manufacturing as well as embellishment techniques.
2. Describe technology involved in the manufacturing processes of home textiles i.e. marker planning, spreading, cutting and sewing techniques as well as embellishment techniques such as embroidery, patchwork, appliqué and quilting.
3. Demonstrate various home textile manufacturing and embellishment techniques.
4. Classify and differentiate various home textile manufacturing and embellishment techniques.
5. Review and revise the knowledge of various home textile manufacturing and embellishment techniques.
6. Development of various home textile products.

# BCT061 Chemical Processing of Textiles – III

**Course Objectives:**

Objective of the course is to learn, understand, identify synthetic dyes, ecofriendly dyes, acid dyes , printing of textiles, nano finishing and colour matching computer operations and apply dyes: acid dyes, metal complex dyes, mordant dyes, printing operations like flat bed printing, rotary screen printing.

**Unit (1):** **Total lectures required = 8**

Chemistry of dyes and chemicals: Classification of dyes according to chemical constitution, features and specification, dye intermediates, manufacturing principle and ecofriendly dyes.

**Unit (2):** **Total lectures required = 8**

**Acid Dyes:**

Generalized formula and classification of acid dyes, Procedure for application of various types of acid dyes to wool & other fibres (e.g. Nylon & Silk) Nylon; mechanism of acid dyeing & dye fibre bond, effect of different process parameters and role of additives in acid dyeing. Fastness properties of acid dyes.

**Chrome Dyes:** Concept of mordants; formation of chromium complexes; Method of application of chrome dyes (i) Chrome mordant process, (ii) After Chrome process, (iii) Meta Chrome process. Brief idea on fastness properties of chrome dyes.

**Metal Complex Dyes:** General formula and structure, classification of metal complex dyes- 1:1 Metal complex dyes & 1:2 Metal complex dyes; Procedure of application of metal complex dyes and mechanism of dyeing fastness; Properties of metal complex dyes.

**Unit (3):** **Total lectures required = 8**

Printing paste composition. Different styles of printing e.g. Direct, Resist and Discharge, Printing process for different fibres with direct dyes, acid dyes, vat dyes, azoic colours, reactive dyes, pigments, disperse dyes etc. Brief idea on function performed by the different pigment auxiliaries/assistants used in the printing paste. Different methods of printing e.g.- screen, flat bed & rotary screen printing machineries and equipments. Common printing faults their causes and remedies.

**Unit (4):** **Total lectures required = 8**

**Functional finishes**

Type of Nano finishes, preparation, application to textiles, self cleaning finish, UV protection, Anti bacterial nano finish, advantages and disadvantages.

**Unit (5):** **Total lectures required = 8**

**Colour Theory and Evaluation**

Theory of colour, quantification of colours, CIE colour system colour difference, whiteness & yellowness in dyes, CIE lab formula, 555 sort. Application of spectrophotometer; Reflectance & Transmittance: K/S Curve, Theory of computer colour matching & recipe prediction.

**References:**

1. Wool Dyeing by D.M. Lewis
2. Chemistry of dyes and principles of dyeing by Dr. V.A. Shenai.
3. Technology of dyeing by Dr. V.A. Shenai
4. Technology of Printing by Dr . V. A. Shenai
5. An introduction to Textile Printing by W Clarke.
6. Textile Printing by L.W.C. Miles.

**Course Outcomes:**

At the end of the course students will be able to:

1. Knowledge of fundamentals of synthetic dye chemistry, acid dyeing, printing, colour matching computer and nano finish.
2. Understanding of technological involvement with appropriate techniques in acid dyes, metal complex dyes, mordant dyes, textile printing colour matching computer and textile nano finishing.
3. Knowledge of application technique of acid dyes, metal complex dyes, mordant dyes, textile printing, colour matching computer and nano finishing process in industry.
4. Capacity to make the chemical processes under control in acid dyes, metal complex dyes, mordant dyes, textile printing, colour matching computer and textile nano finishing.
5. Understanding for doing project work and aptitude for higher studies in the field of acid dyes, metal complex dyes, mordant dyes, computer colour matching and nano finishes.

# BCT062 Knitting and Nonwovens

**Course Objectives:**

To label, classify, explain, illustrate, characterize, evaluate and sustainable development of warp and weft knitted fabric structures, their manufacturing machines and techniques, knitted garments, various nonwoven fabric application and its manufacturing techniques.

**Unit (1):** **Total lectures required = 8**

Introduction to knitting:
Essential features of knitted fabric, advantages and limitations with reference to weaving knitted loop structure, terms and definitions.
Knitting needles and cycle of knitting.
Knitted garments : designing and specifications

**Unit (2):** **Total lectures required = 8**

Weft Knitting machine:
Types of weft knitting m/c: flat and circular. Essential parts of circular weft knitting machine.
Flat weft knitting machine.
Weft knitting machines for plain knit, rib, interlock and purl knitting machines, variation for tuck and miss stitches.
production calculation of weft knitting machines.
Technical specification of modern weft knitting machines.

**Unit (3):** **Total lectures required = 8**

**Warp knitiing machine:**

Basic warp knitted structures, under lap and overlap.
Essential parts of warp knitting m/c,
Brief idea of the Working principal of Tricot, Rachel & crochet Machines
calculation of Production pertaining to the warp knitting machine
Technical specifications of modern warp knitting machines

**Unit (4):** **Total lectures required = 8**

Non wovens – classification and brief idea of various types of nonwoven fabric,
non-woven fabric manufacturing process,
Fibres used, Principles of web formation,
bonding techniques.
Various finishing used for non woven.

**Unit (5):** **Total lectures required = 8**

Principles of needle punching, Process variables and their effect on properties of needle punched fabrics.
Brief idea of Stitch bonded fabrics, their manufacture and properties.
Brief idea of spun bonded fabrics production
Technical specifications of modern nonwoven machines.

**References:**

1. Non woven by P Madhavanmoorthi
2. Knitting Technology David J Spencer Knitting Technology Ajgaonkar
3. Warp Knitting Production S Ray, Melliand
4. Manual of Nonwoven R Krcma
5. Nonwoven Fabric Technology M Mc Donald
6. Nonwoven technology by Milin patel & Dhruvkumar bramahat.

**Course outcome:**

At the end of the course students will be able to:

1. Recall the knowledge in warp and weft knitted fabric structures, their manufacturing machines and techniques, knitted garments, various nonwoven fabric application and its manufacturing techniques.
2. Classify and identify warp and weft knitted fabric structures, their manufacturing machines and techniques, knitted garments, various nonwoven fabric application and its manufacturing techniques.
3. Illustrate and show warp and weft knitted fabric structures, their manufacturing machines and techniques, knitted garments, various nonwoven fabric application and its manufacturing techniques.
4. Manage and investigate warp and weft knitted fabric structures, their manufacturing machines and techniques, knitted garments, various nonwoven fabric application and its manufacturing techniques.
5. Determine and evaluate warp and weft knitted fabric structures, their manufacturing machines and techniques, knitted garments, various nonwoven fabric application and its manufacturing techniques.
6. Create and develop warp and weft knitted fabric structures, their manufacturing machines and techniques, knitted garments, various nonwoven fabric application and its manufacturing techniques.

# BCT063 Fibres for Carpets & Textiles

**Course Objectives:**

A comprehensive understanding of both natural and manmade fibers, their production processes and their applications

**Unit (1): Total lectures required = 8**

Different vegetable fibers (Kapok, Jute, Flax, Hemp, Ramie, Sisal, Pineapple, Coir) and manmade fibers (polyester, nylon 6, nylon 66 and polypropylene), Melt spinning: Raw materials of polyester, nylon 6, nylon 66 and polypropylene, technology of polymerization and extrusion of polyester, nylon 6, nylon 66 and polypropylene.

**Unit (2): Total lectures required = 8**

Solution spinning: Dope Preparation and spinning of cellulose acetate, cellulose diacetate and triacetate, Dry spinning and wet spinning of acrylic. Significance and types of co-monomers used during polymerization of acrylic, Process flow diagram for solution wet spinning of viscose rayon. Chemistry of viscose rayon formation, Influence of various additives and temperature of the regeneration bath and their influence on the process and properties of viscose rayon, Various types of rayons, i.e. high wet modulus, TenX, etc.

**Unit (3): Total lectures required = 8**

Drawing and heat setting of fibers: Objective of drawing. Concept of neck drawing, Effect of drawing conditions on the structure and properties of fiber. Objective of heat setting. Effect of heat setting parameters on the structure and properties of fiber. Spin finish: Objective, properties and application of spin finish.

**Unit (4): Total lectures required = 8**

Introduction to Manmade Fibers: Fundamentals of high performance fibers such as Nomex, Kevlar, Carbon, Spandex, etc. their properties and applications. Developments in fibers production such as micro denier, bicomponent, hollow fibers, etc. Sustainability aspects of manmade fibers.

**Unit (5): Total lectures required = 8**

Introduction to texturing General principles involved in the manufacture of textured yarns Bulked yarns Mechanisms of setting and texturing Thermo-mechanical texturing, draw texturing, air jet texturing

**References:**


1. Vaidya A A, “Production of Synthetic Fibers”, 1st Ed., Prentice Hall of India, New Delhi, 1988.
2. “Manufactured Fiber Technology”, 1st Ed. V B Gupta and V K Kothari, 1st Ed., Chapman and Hall, London, 1997.
3. Mark H F, Atlas S M, Cernia E, “Man Made Fibre Science and Technology”, 1st Ed.,Vol.1, 2, 3, Science Publishers, New York, 1967.
4. Macintyre J E, “Synthetic Fibres”, Wood head Fiber Science Series, UK, 2003.
5. “Hand Book of Fibre Chemistry”, Ed. M Lewin and E M Pearce, Mercel Dekker Inc., 1998.
6. A Textbook Of Fibre Science And Technology : Mishra, S.P.

**Course outcome:**

At the end of this course students will be able to

1. list various types of vegetable and manmade fibres.
2. explain the production process of manmade fibres.
3. demonstrate the application of manmade fibres in different industries.
4. compare the properties of manmade fibres with natural fibres.
5. assess the environmental impact and sustainability of manmade fibres.
6. design innovative uses for manmade fibres.

# LAB 4: BCT651 Carpet & Textile Testing II Lab

**Course Objectives**

1. To get aware of principles involved in tensile, tear and bursting strength testing of textiles
2. To get an understanding of the measurement of functional properties, such as flame proofness & water proofness of textile materials
3. To get aware of measurement of serviceability aspects of textile materials
4. To learn about measurement of durability and appearance retention properties of carpet.

**Experiments**

1. Determination of C.S.P. of yarn using Lea strength tester.
2. Comparison of grams per square meter of a piece of fabric (both by theoretical calculations and practical measurements).
3. Determination of ends/inch; picks/inch; warp & weft count from a given piece of fabrics.
4. Determination of bending length, flexural rigidity & bending modulus using Shirley fabric stiffness tester.
5. Determination of abrasion resistance and pilling resistance using Martindale Abrasion Tester.
6. Determination of fabric tensile strength using universal tester.
7. Determination of flammability of a fabric.
8. Determination of water repellency using Bundesmann water repellency tester.
9. Determination of pilling of fabric by pill box.
10. Determination of fabric tear strength.
11. Determination of thickness loss under dynamic loading.
12. Determination of thickness loss under static loading.
13. Determination of Abrasion resistance of carpet and weight loss.
14. Determination of degree of appearance retention of a carpet using Hexapod Tumbler Tester.

NOTE: Experiments shall be decided on factors like:

• Facilities installed at Institute. • Accessibility to Industry & nearby Institutes. • Trend of Technological Developments in National & International perspective.

**Course outcome**

After completion of this course, students will be able to:

1. Appreciate the importance of mechanical strength testing of Textile materials and its measurement using various instruments.
2. Appreciate the importance of measurement of working life/durability/surface appearance of textile articles.
3. Appreciate the importance of measurement of various functional properties of textile articles.
4. Understand the factors responsible for appearance retention and durability aspects of carpet.

# LAB 5: BCT651 Carpet & Textile Design Lab

**Course Objectives**

1. To learn about various CAD tools for manufacturing of carpet designs.
2. To prepare various structures of design like geometrical, natural etc.
3. To prepare various designs of fabrics, dhurries and carpets.
4. To forecast colour as per market trends.

**Experiments**

1. Introduction / CAD practical with help of various tools/ Quality calculation on Software.
2. Wool Consumption of Rugs through Software.
3. Geometrical structures, Exploration of forms, Shapes & line with in the natural forms or objects. Still Life drawings.
4. Stripes & Checks effect on fabric,
5. Colour & Design Creations: Durries - 30ct, 60ct, 80ct And Boxes & Round Compositions, Concept of shade tone – Chromo tone/ ARS.
6. Carpet designs: Tufted - Floral & Modern Designs, Tibbetan – Modern geometrical & Floral.

a. Converting Natural form of designs into Abstract, Modern & Contemporary.

b. Persian/Traditional Designs/Erase Pattern Design.

7. Colour Forecasting/ Colour Trends Carpet & Textiles.

**Course outcome**

At the end of course students will be able to:

1. Demonstrate various CAD tools.
2. Preparation of various designs forms like geometrical, natural etc.
3. Preparation of various designs of fabrics, dhurries & carpets.
4. Forecast colour as per market trends with the help of forecast series.

# LAB 6: BCT651 Home Textile Technology Lab

**Course objectives**

1. To learn about sewing machines and practice on it.
2. To prepare various samples of temporary and permanent stitches.
3. To prepare various samples necessary for construction of home textiles.
4. To prepare various samples necessary for embellishment of home textiles.

**Experiments**

1. To study the various parts and its functions of the lock stitch Sewing Machines.
2. Practice of machine stitch on paper.
3. Practice of machine stitch on fabric.
4. Sample preparation for basic Hand Stitches.

i) Temporary Stitches (Even basting, Uneven Basting, Diagonal, Slip stitch)
ii) Permanent Stitches (Running, Hemming, Back, Run and Back, Over casting, Whipping)

5. Preparation of seam samples.

i) Superimposed seam.
ii) Lapped seam.
iii) Bound seam.
iv) Ornamental seam.
v) Edge finishing.
vi) Flat seam.

6. Sample preparation for seam finishes and self-finish seams.

i. Seam finishes (Clean finish, Bias binding, Bound finish, Hand overcast).
ii. Self finished seam (Standing fell, Drapery French, Single needle, Quick flat seam, Lap seam)

7. Preparation of samples for following fullness tools: -

a. Darts.
b. Tucks.
c. Pleats.
d. Godets.
e. Gathers.
f. Shirrs.
g. Frills.

8. Preparation of samples for different types of Plackets.
9. Sample preparation for hand embroidery stitches.

(Running, Stem, Single lace, Double lace, Chain, Satin, Lock, Whip, Cross, Pipe, Loop, Flat, Knotted).

**Course outcomes**

At the end of the course students will be able to:

1. Identify various parts of lock stitch sewing machine.
2. Operate various sewing machines.
3. Construction of various samples of basic hand stitches, seams, seam finishes and self-finish seams, which are necessary for construction of home textiles.
4. Construction of various samples of fullness tools, plackets and hand embroidery stitches, which are necessary for embellishment of home textiles.